INTEGRAL SCHEME COVERS GENERICALLY FINITE-ETALE AND LOCALLY REFINING A PRESCRIBED ETALE COVER

THE STACKS PROJECT AUTHORS

Editorial note. At thed=1 base step the author has already shownX is a scheme. The integral identity coverY=X satisfies the conclusion and locally factors through the open covering; the printedY=V there is an immediate variable-choice slip because a general open cover need not be integral. Original text retained.

Editorial note. The phrase finite overV_j in the normalization base-change paragraph can be read as integral/affine, which is all that the scheme and quasi-compactness conclusion uses. No finiteness of normalization over an arbitrary base is added.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Quasi-compact quasi-separated algebraic spaces, dense open scheme loci, relative normalization, quasi-finite separated maps, etale diagonal sections and fibre-cardinality stratification. The bounded-fibre argument is retained author support.

Source reference. Original lemma 03JS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4650–4650. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03JU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4652–4652. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03JV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4653–4653. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03JT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4651–4651. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03IH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4603–4603. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 03W7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5102–5102. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03JA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4632–4632. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03JO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4646–4646. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03JN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4645–4645. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03JM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4644–4644. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03JQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4648–4648. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 09YB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12559–12559. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 06NH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8536–8536. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02NW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3495–3495. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0ABS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13030–13030. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0ABP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13027–13027. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0825: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10240–10240. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0823: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10238–10238. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0831: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10271–10271. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 088G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10461–10461. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03KO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4684–4684. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07S8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9893–9893. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03ZR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5230–5230. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

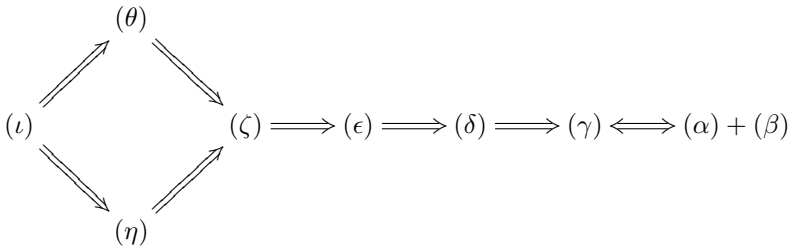
Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S such that $|X|$ has finitely many irreducible components.*

1. *There exists a surjective integral morphism $Y \to X$ where Y is a scheme such that f is finite étale over a quasi-compact dense open $U \subset X$,*
2. *given a surjective étale morphism $V \to X$ we may choose $Y \to X$ such that for every $y \in Y$ there is an open neighbourhood $W \subset Y$ such that $W \to X$ factors through V.*

Lemma 0.2. *Let S be a scheme. Let X be an algebraic space over S. Consider the following conditions on X:*

(α) *For every $x \in |X|$, the equivalent conditions of Lemma 03JS hold.*
(β) *For every $x \in |X|$, the equivalent conditions of Lemma 03JU hold.*
(γ) *For every $x \in |X|$, the equivalent conditions of Lemma 03JV hold.*
(δ) *The equivalent conditions of Lemma 03JT hold.*
(ϵ) *The equivalent conditions of Lemma 03IH hold.*
(ζ) *The space X is Zariski locally quasi-separated.*
(η) *The space X is quasi-separated*
(θ) *The space X is representable, i.e., X is a scheme.*
(ι) *The space X is a quasi-separated scheme.*

We have



Proof. The implication $(\gamma) \Leftrightarrow (\alpha) + (\beta)$ is immediate. The implications in the diamond on the left are clear from the definitions.

Assume (ζ), i.e., that X is Zariski locally quasi-separated. Then (ϵ) holds by Properties of Spaces, Lemma 03W7.

Assume (ϵ). By Lemma 03IH there exists a Zariski open covering $X = \bigcup X_i$ such that for each i there exists a scheme U_i and a quasi-compact surjective étale morphism $U_i \to X_i$. Choose an i and an affine open subscheme $W \subset U_i$. It suffices to show that $W \to X$ has universally bounded fibres, since then the family of all these morphisms $W \to X$ covers X. To do this we consider the diagram

$$\begin{array}{ccc} W \times_X U_i & \xrightarrow{p} & U_i \\ {\scriptstyle q}\downarrow & & \downarrow \\ W & \longrightarrow & X \end{array}$$

Since $W \to X$ factors through X_i we see that $W \times_X U_i = W \times_{X_i} U_i$, and hence q is quasi-compact. Since W is affine this implies that the scheme $W \times_X U_i$ is quasi-compact. Thus we may apply Morphisms, Lemma 03JA and we conclude that p has universally bounded fibres. From Lemma 03JO we conclude that $W \to X$ has universally bounded fibres as well.

Assume (δ). Let U be an affine scheme, and let $U \to X$ be an étale morphism. By assumption the fibres of the morphism $U \to X$ are universally bounded. Thus also the fibres of both projections $R = U \times_X U \to U$ are universally bounded, see Lemma 03JN. And by Lemma 03JM also the fibres of $R \to X$ are universally bounded. Hence for any $x \in X$ the fibres of $|U| \to |X|$ and $|R| \to |X|$ over x are finite, see Lemma 03JQ. In other words, the equivalent conditions of Lemma 03JV hold. This proves that $(\delta) \Rightarrow (\gamma)$. □

Proof. The proof is the (roughly) same as the proof of Lemma 09YB with additional technical comments to obtain the dense quasi-compact open U (and unfortunately changes in notation to keep track of U).

Part (1) is the special case of part (2) where $V = X$.

Proof of (2). Choose a surjective étale morphism $V' \to V$ where V' is a scheme. It is clear that we may replace V by V' and hence we may assume V is a scheme. Since X is quasi-compact, there exist finitely many affine opens $V_i \subset V$ such that $V' = \coprod V_i \to X$ is surjective. After replacing V by V' again, we see that we may assume V is affine. Since X is quasi-separated, hence reasonable, there exists an integer d bounding the degree of the geometric fibres of $V \to X$ (see Lemma 0.2).

By induction on $d \geq 1$ we will prove the following induction hypothesis (H_d):

- for any quasi-compact and quasi-separated algebraic space X with finitely many irreducible components, for any $m \geq 0$, for any quasi-compact and separated schemes V_j, $j = 1, \ldots, m$, for any étale morphisms $\varphi_j : V_j \to X$, $j = 1, \ldots, m$ such that d bounds the degree of the geometric fibres of $\varphi_j : V_j \to X$ and $\varphi = \coprod \varphi_j : V = \coprod V_j \to X$ is surjective, the statement of the lemma holds for $\varphi : V \to X$.

If $d = 1$, then each φ_j is an open immersion. Hence X is a scheme and the result holds with $Y = V$. Assume $d > 1$, assume (H_{d-1}) and let m, $\varphi : V_j \to X$, $j = 1, \ldots, m$ be as in (H_d).

Let $\eta_1, \ldots, \eta_n \in |X|$ be the generic points of the irreducible components of $|X|$. By Properties of Spaces, Proposition 06NH there is an open subscheme $U \subset X$ with $\eta_1, \ldots, \eta_n \in U$. By shrinking U we may assume U affine and by Morphisms, Lemma 02NW we may assume each $\varphi_j : V_j \to X$ is finite étale over U. Of course, we see that U is quasi-compact and dense in X and that $\varphi_j^{-1}(U)$ is dense in V_j. In particular each V_j has finitely many irreducible components.

Fix $j \in \{1, \ldots, m\}$. As in Morphisms of Spaces, Lemma 0ABS we let Y_j be the normalization of X in V_j. We obtain a factorization

$$\begin{array}{ccc} V_j & \longrightarrow & Y_j \\ & {\scriptstyle \varphi_j} \searrow \quad \swarrow {\scriptstyle \pi_j} & \\ & X & \end{array}$$

with π_j integral and $V_j \to Y_j$ a quasi-compact open immersion. Since Y_j is the normalization of X in V_j, we see from Morphisms of Spaces, Lemmas 0ABP and 0825 that $\varphi_j^{-1}(U) \to \pi_j^{-1}(U)$ is an isomorphism. Thus π_j is finite étale over U. Observe that V_j is scheme theoretically dense in Y_j because Y_j is the normalization of X in V_j (follows from the characterization of relative normalization in Morphisms of Spaces, Lemma 0823). Since V_j is quasi-compact we see that $|V_j| \subset |Y_j|$ is dense, see Morphisms of Spaces, Section 0831 (and especially Morphisms of Spaces, Lemma 088G). It follows that $|Y_j|$ has finitely many irreducible components. Then $V_j \times_X Y_j$ is a quasi-compact, separated scheme (being finite over V_j) and

$$V_j \times_X Y_j = V_j \amalg W_j$$

Here the first summand is the image of $V_j \to V_j \times_X Y_j$ (which is closed by Morphisms of Spaces, Lemma 03KO and open because it is étale as a morphism between algebraic spaces étale over Y) and the second summand is the (open and closed) complement.

The étale morphism $W_j \to Y_j$ has geometric fibres of cardinality $< d$. Namely, this is clear for geometric points of $V_j \subset Y_j$ by inspection. Since $|V_j| \subset |Y_j|$ is dense, it holds for all geometric points of Y_j by Lemma 07S8 (the degree of the fibres of a quasi-compact étale morphism does not go up under specialization). By (H_{d-1}) applied to $V_j \amalg W_j \to Y_j$ we find a surjective integral morphism $Y_j' \to Y_j$ with Y_j' a scheme, which Zariski locally factors through $V_j \amalg W_j$, and which is finite étale over a quasi-compact dense open $U_j \subset Y_j$. After shrinking U we may and

do assume that $\pi_j^{-1}(U) \subset U_j$ (we may and do choose the same U for all j; some details omitted).

We claim that

$$Y = \coprod\nolimits_{j=1,\ldots,m} Y_j' \longrightarrow X$$

is the solution to our problem. First, this morphism is integral as on each summand we have the composition $Y_j' \to Y \to X$ of integral morphisms (Morphisms of Spaces, Lemma 03ZR). Second, this morphism Zariski locally factors through $V = \coprod V_j$ because we saw above that each $Y_j' \to Y_j$ factors Zariski locally through $V_j \amalg W_j = V_j \times_X Y_j$. Finally, since both $Y_j' \to Y_j$ and $Y_j \to X$ are finite étale over U, so is the composition. This finishes the proof. □

References